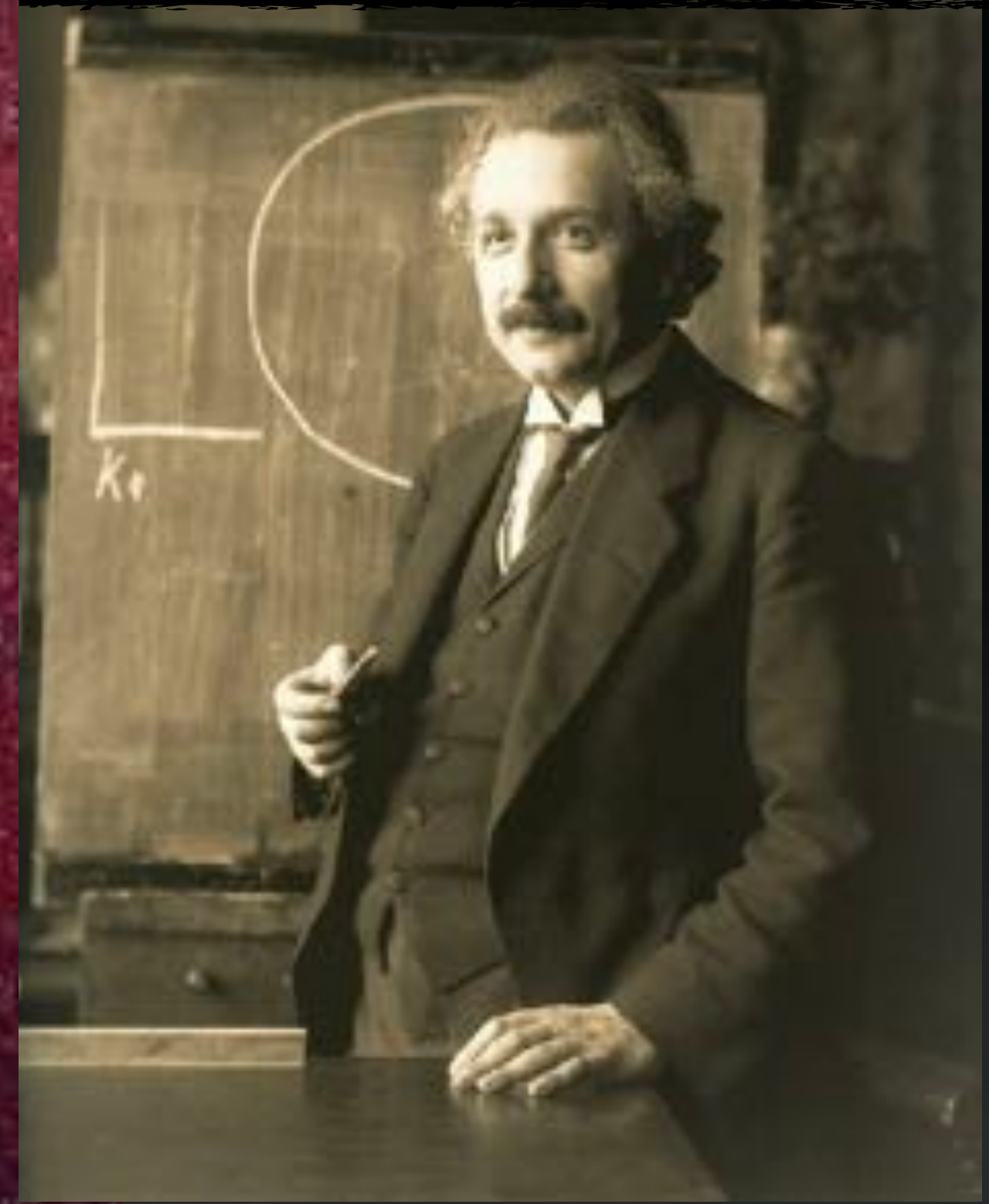
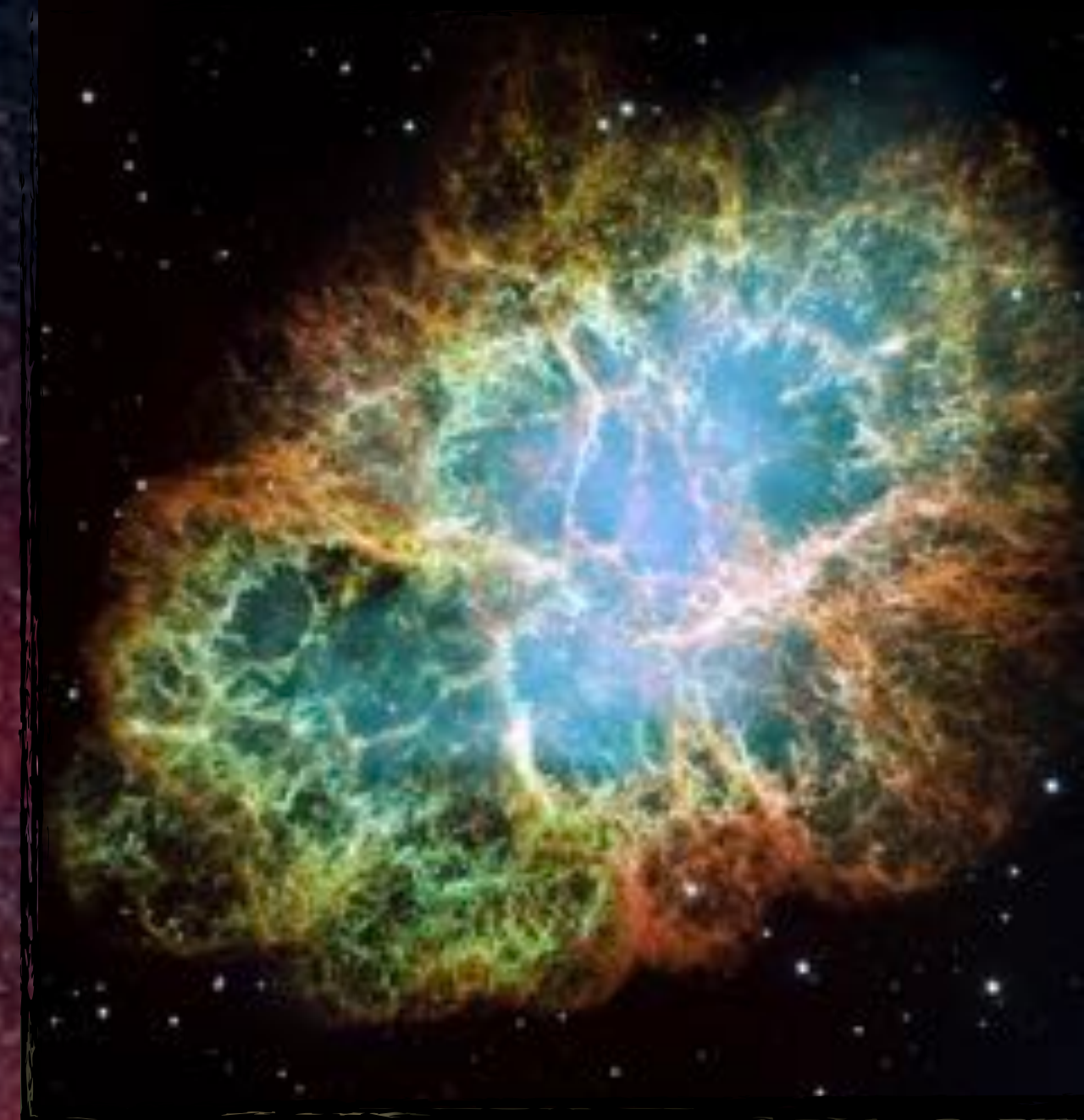
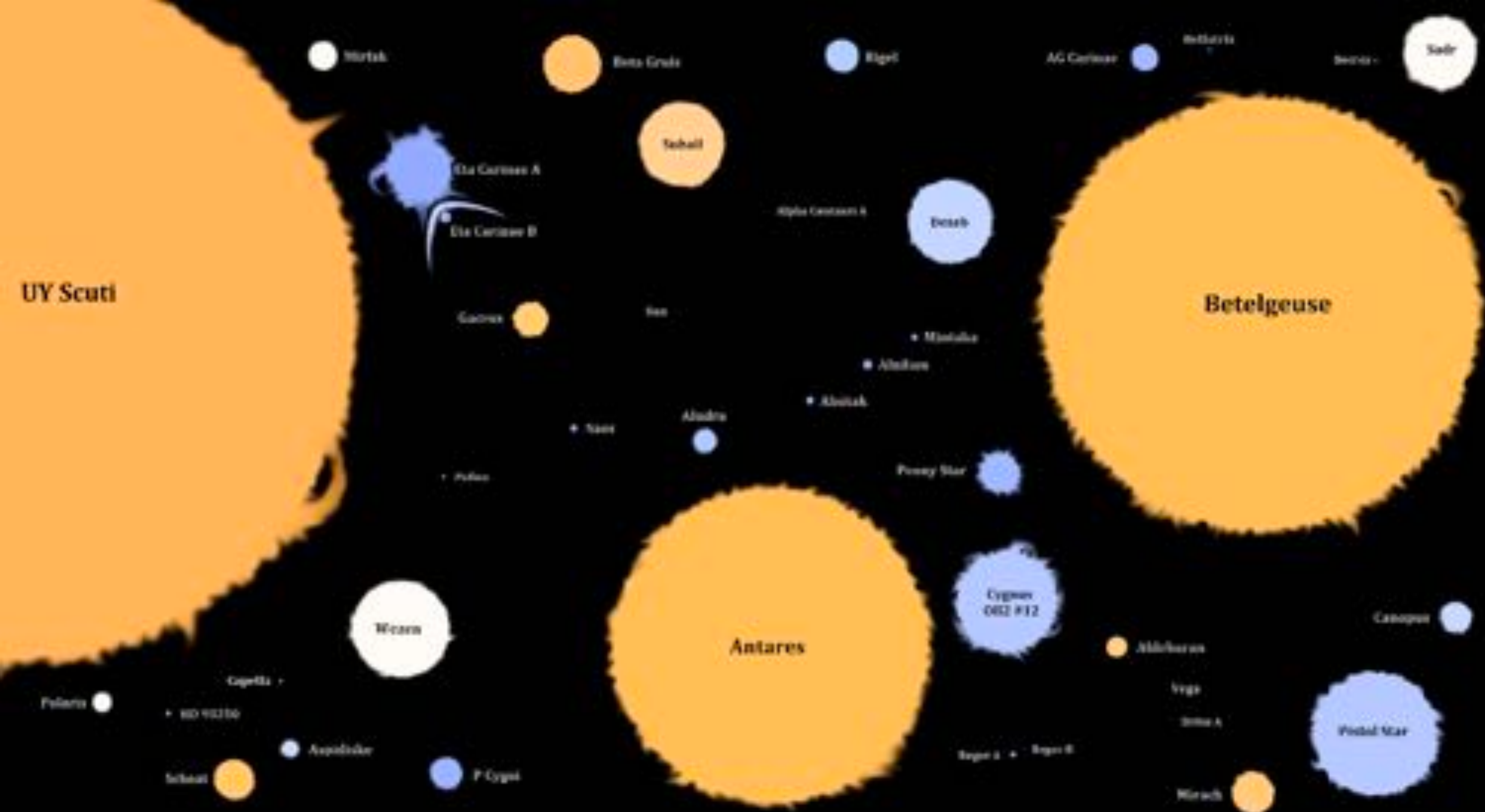
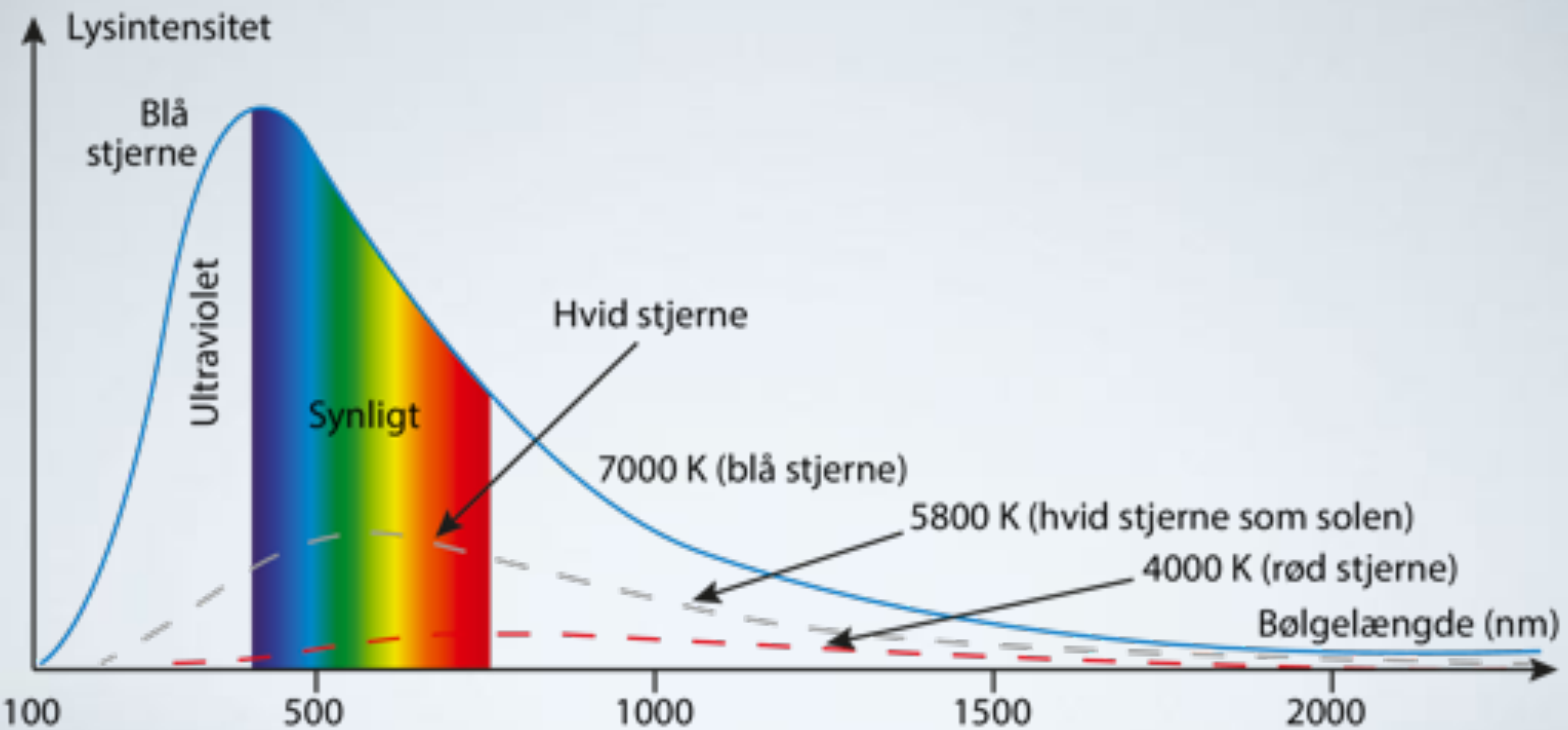






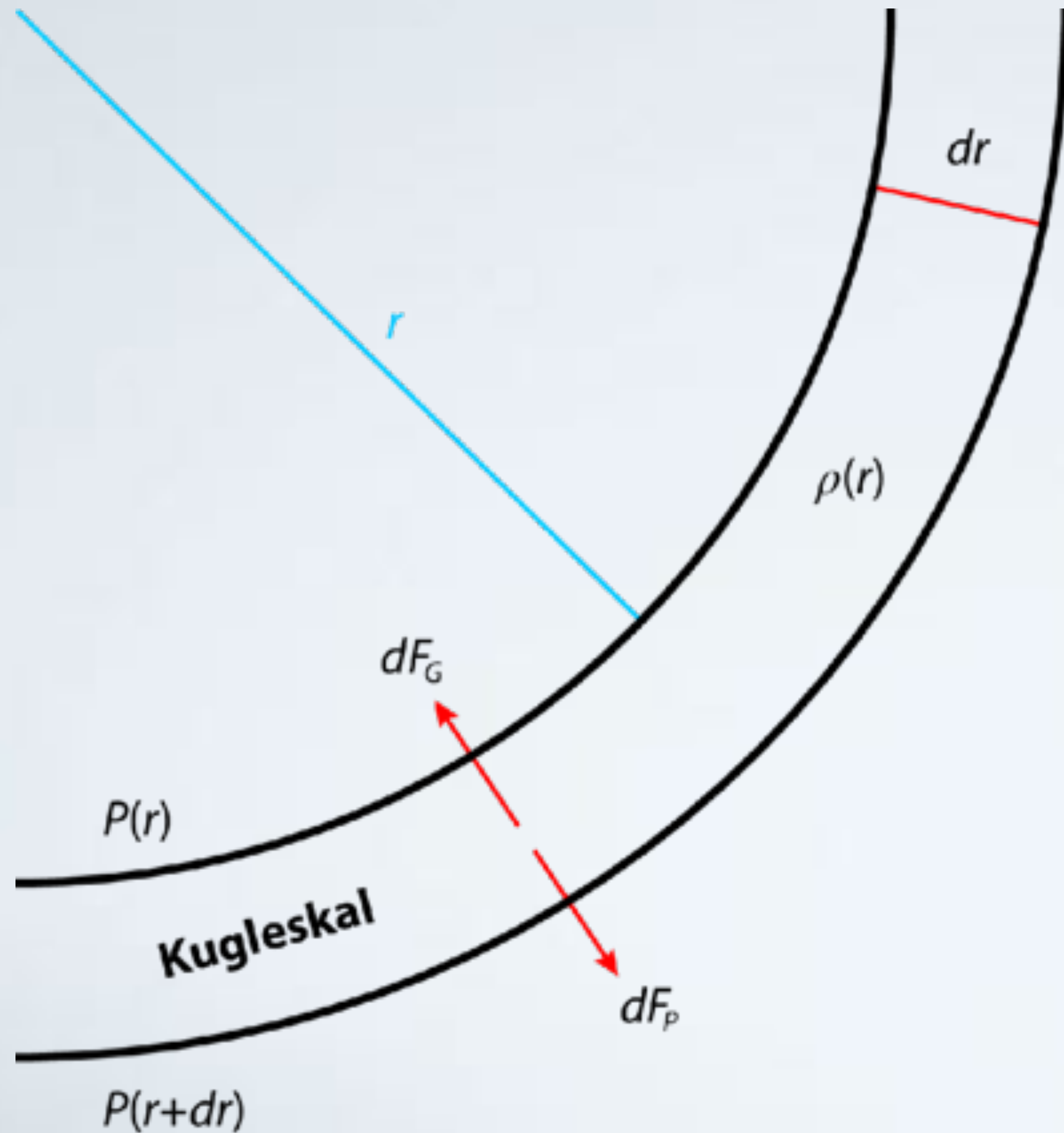


HYDROSTATISK LIGEVÆGT

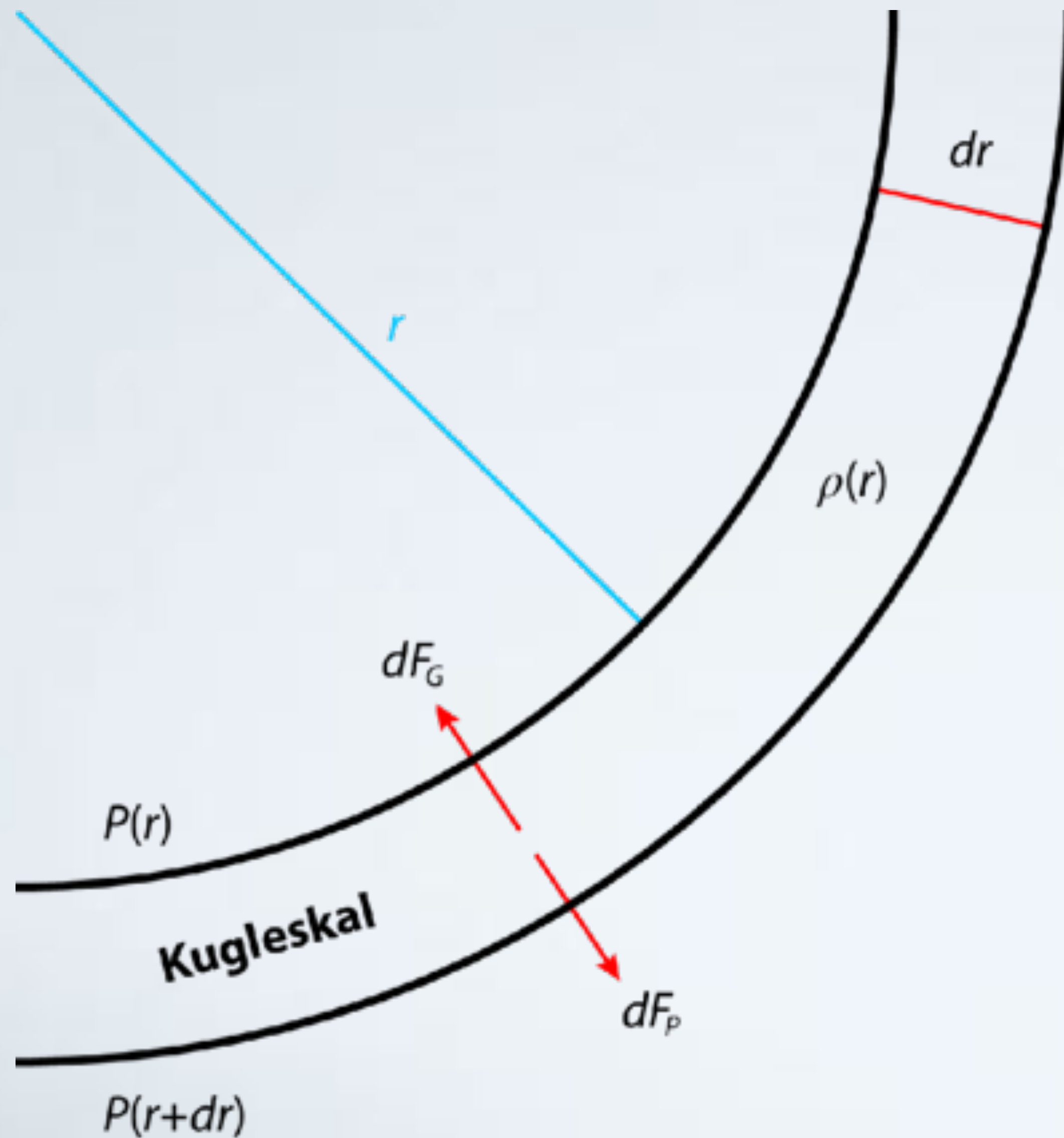
SKALSÆTNINGEN

- Kraften på en partikel med masse m fra en sfærisk symmetrisk massefordeling med total masse M er den samme som en punkt masse med masse M placeret i centrum af massefordelingen.
- Kraften på en partikel med masse m der befinder sig i en sfærisk symmetrisk skal med masse M er 0.

HYDROSTATISK LIGEVÆGT

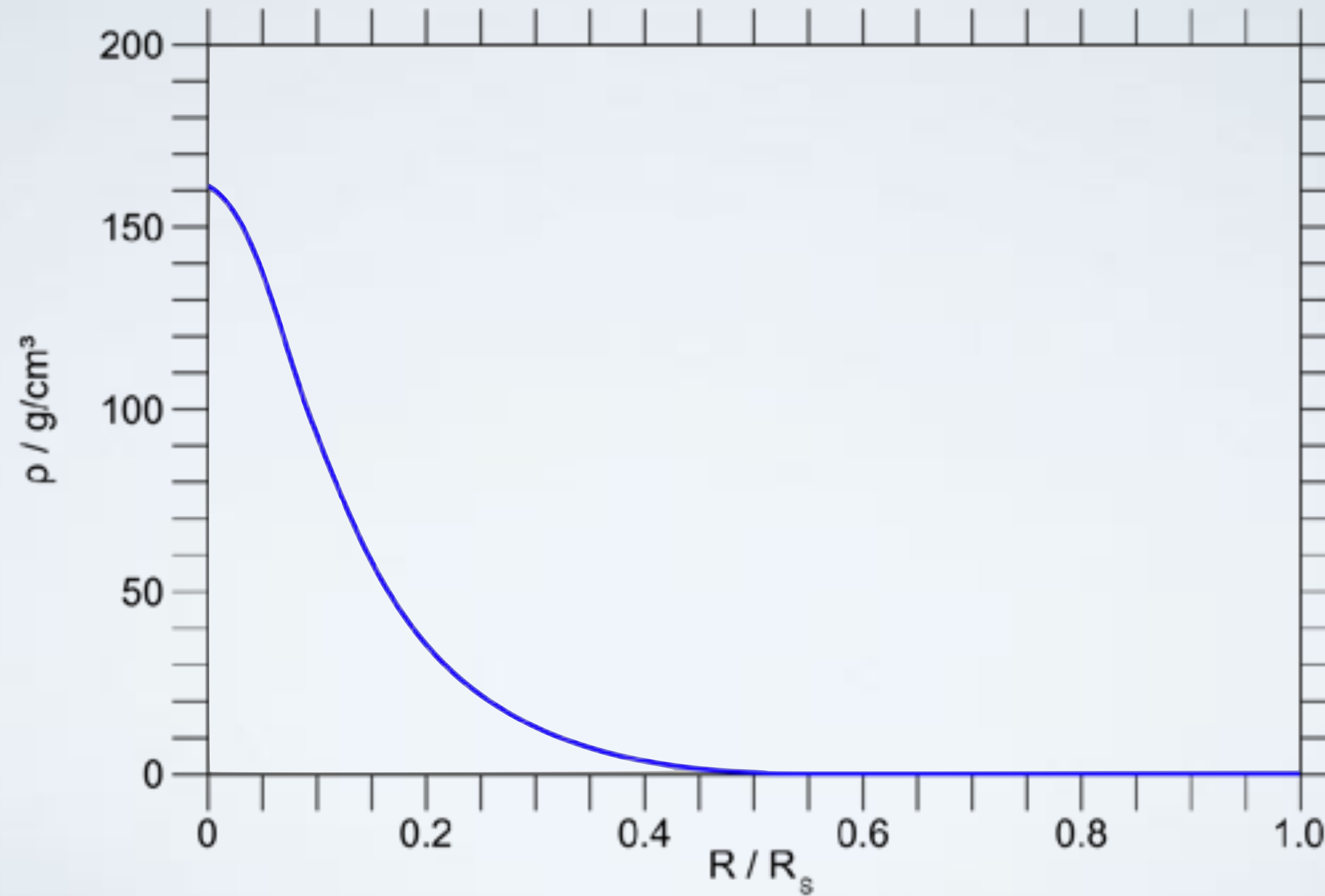


HYDROSTATISK LIGEVÆGT



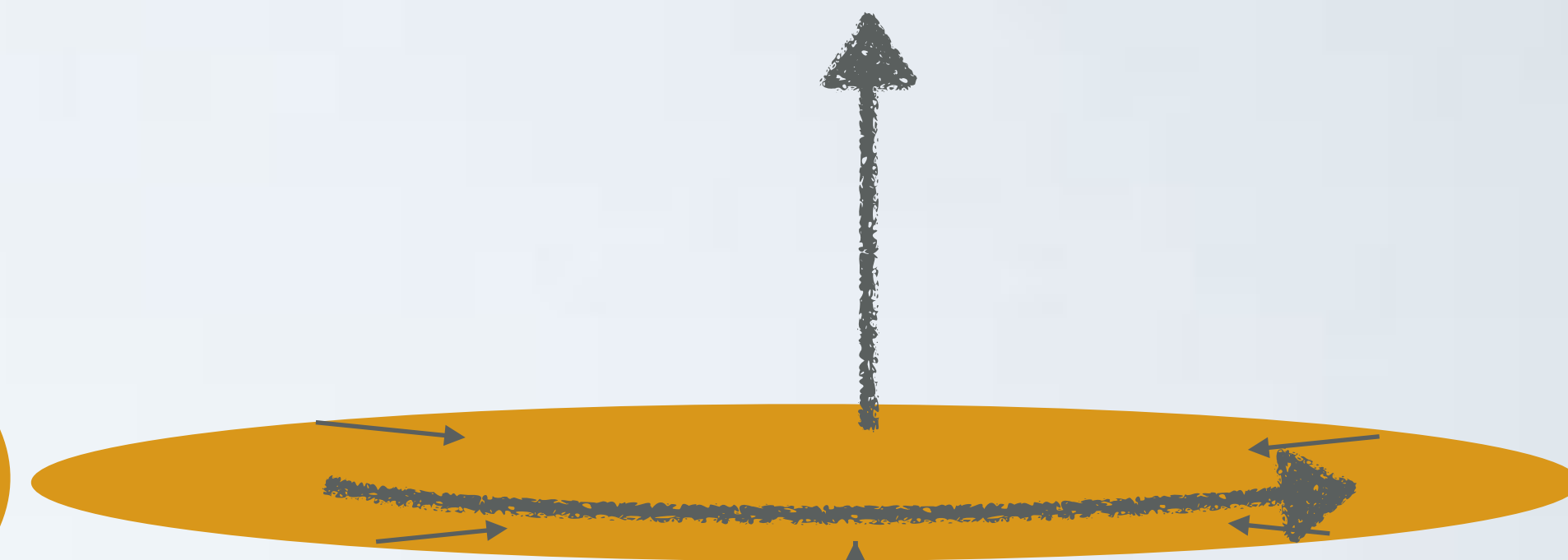
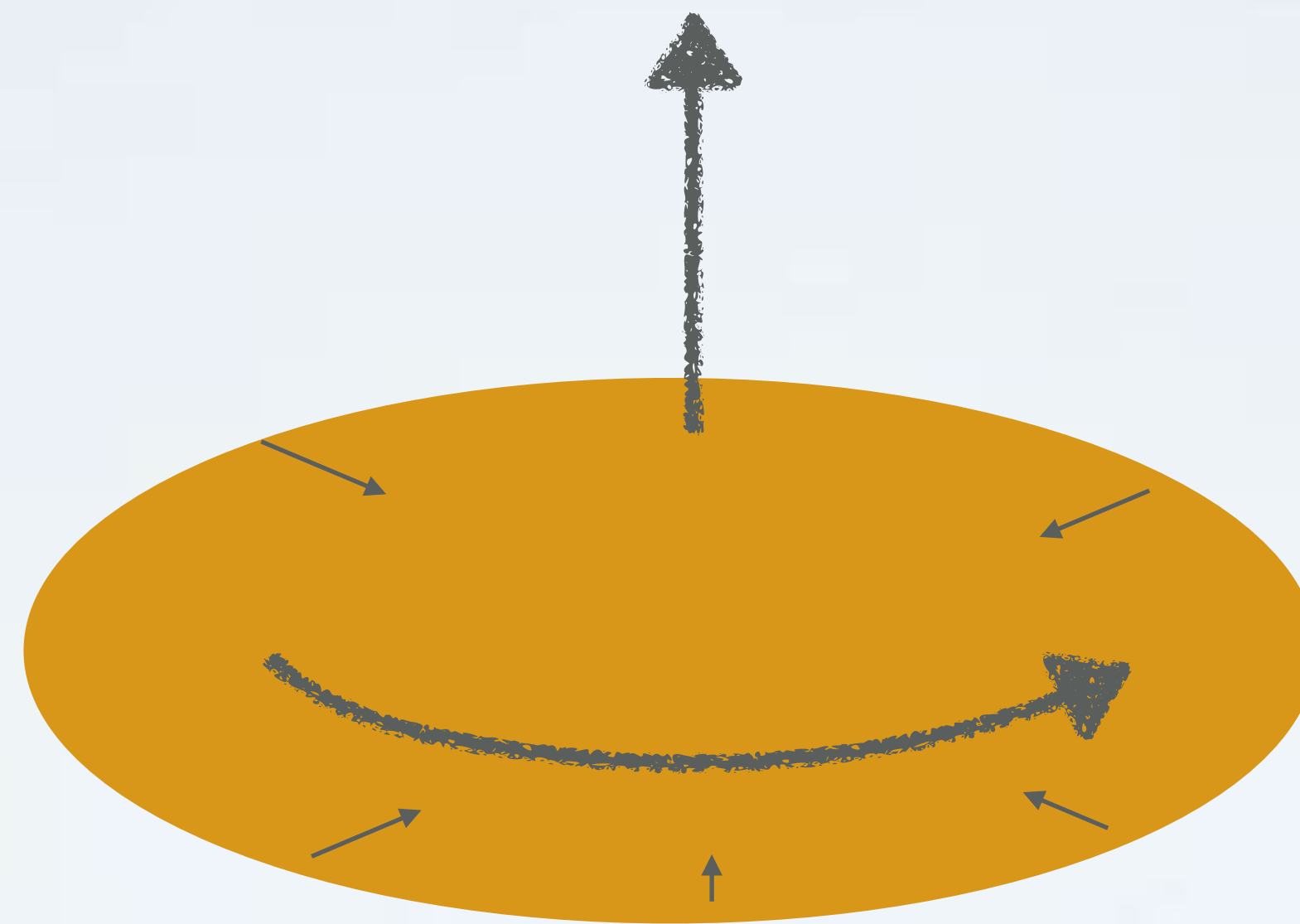
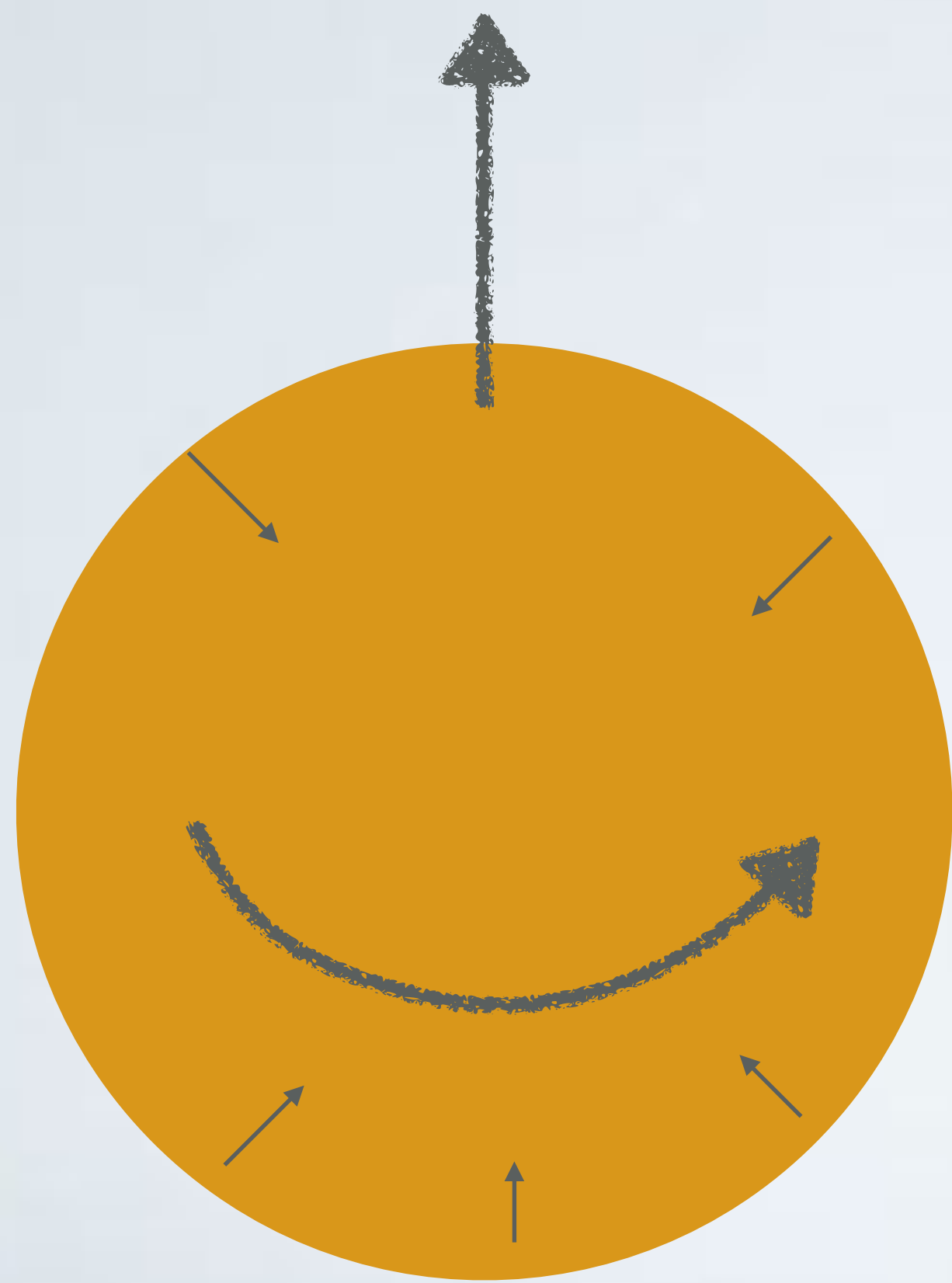
$$\frac{dP}{dr} = - \frac{GM(r)\rho(r)}{r^2}$$
$$\frac{dM}{dr} = 4\pi r^2 \rho(r)$$

EKSEMPEL: SOLEN



STJERNER

STJERNEDANNELSE



HVOR KOMMER TRYKKET FRA?

1. Idealgas ligningen:

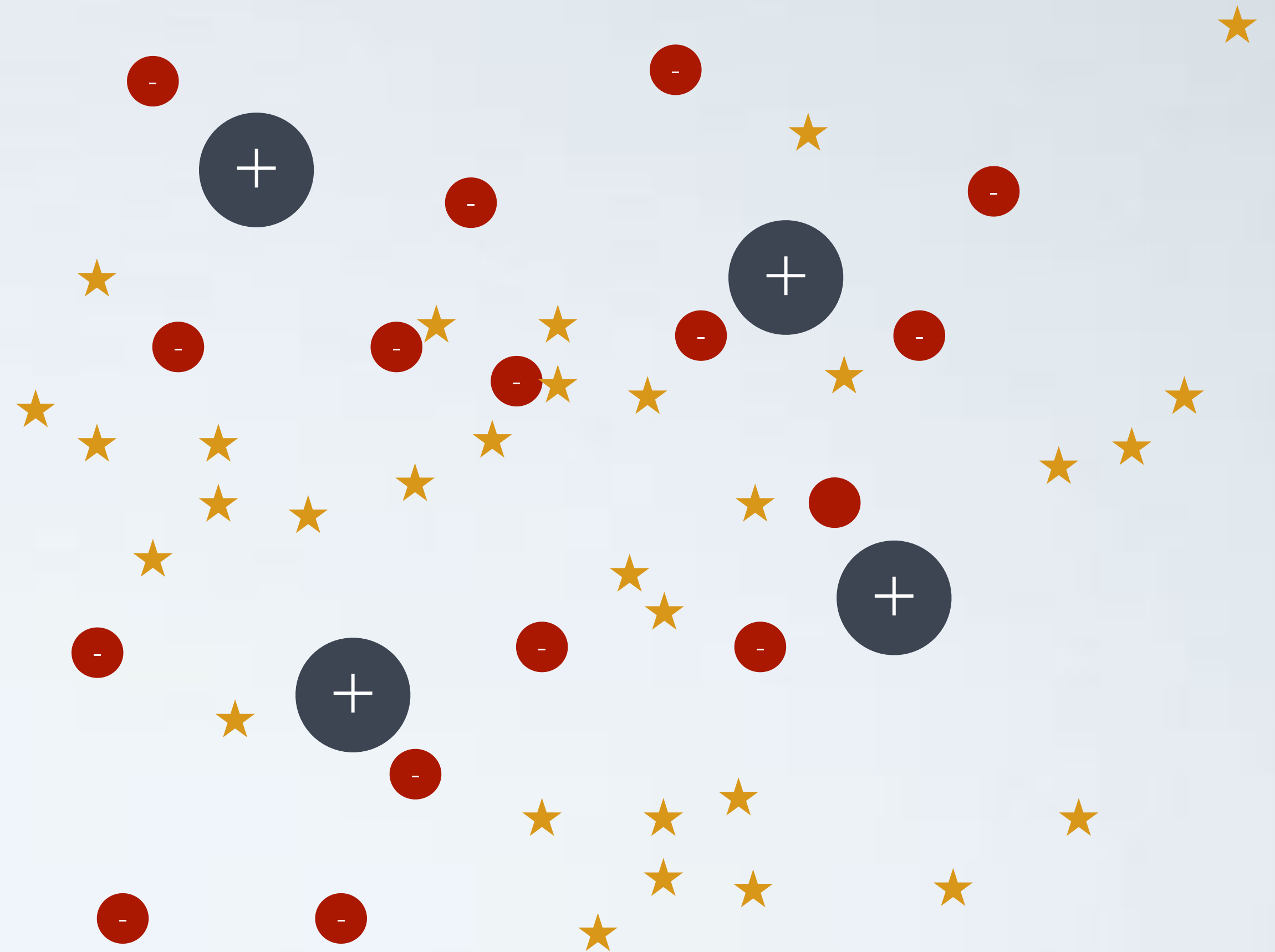
$$P(r) = \frac{k_B \rho(r) T(r)}{\mu m_p}$$

2. Strålingstryk fra fotonerne:

$$P(r) = \frac{\pi^2 k_B^4 T(r)^4}{45 (\hbar c)^3}$$

3. Kvantemekanisk tryk:

$$P_{\text{deg}}(r) = \begin{cases} (3\pi^2)^{\frac{2}{3}} \frac{\hbar^2}{5m} \left(\frac{N_{e,n}}{V} \right)^{\frac{5}{3}} \\ \frac{1}{4} (3\pi^2)^{\frac{1}{3}} \hbar c \left(\frac{N_{e,n}}{V} \right)^{\frac{4}{3}} \end{cases}$$



ikke-relativistisk

ultra-relativistisk

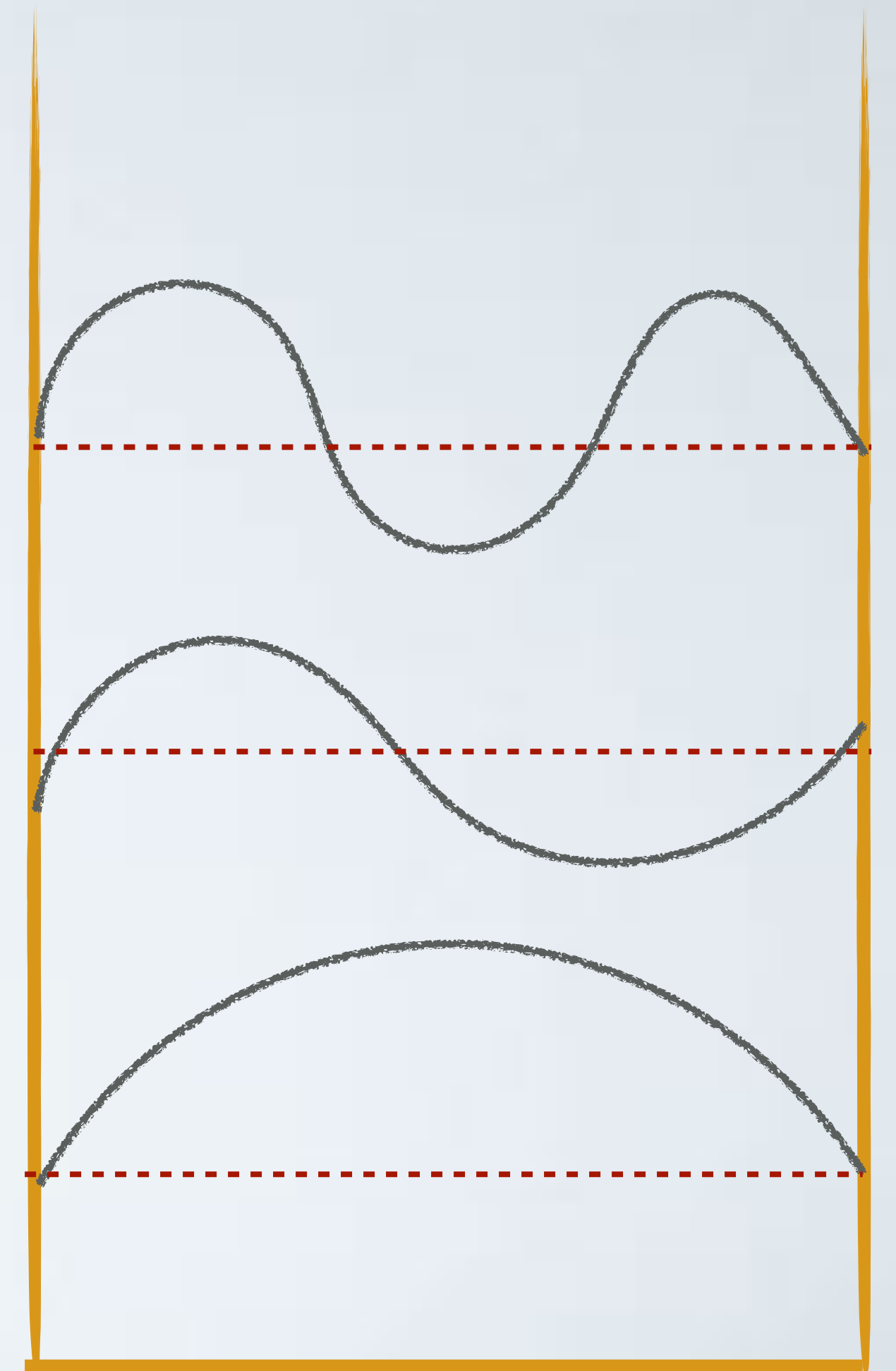
KVANTEMEKANISK TRYK

- Heisenbergs ubestemthedsrelation:

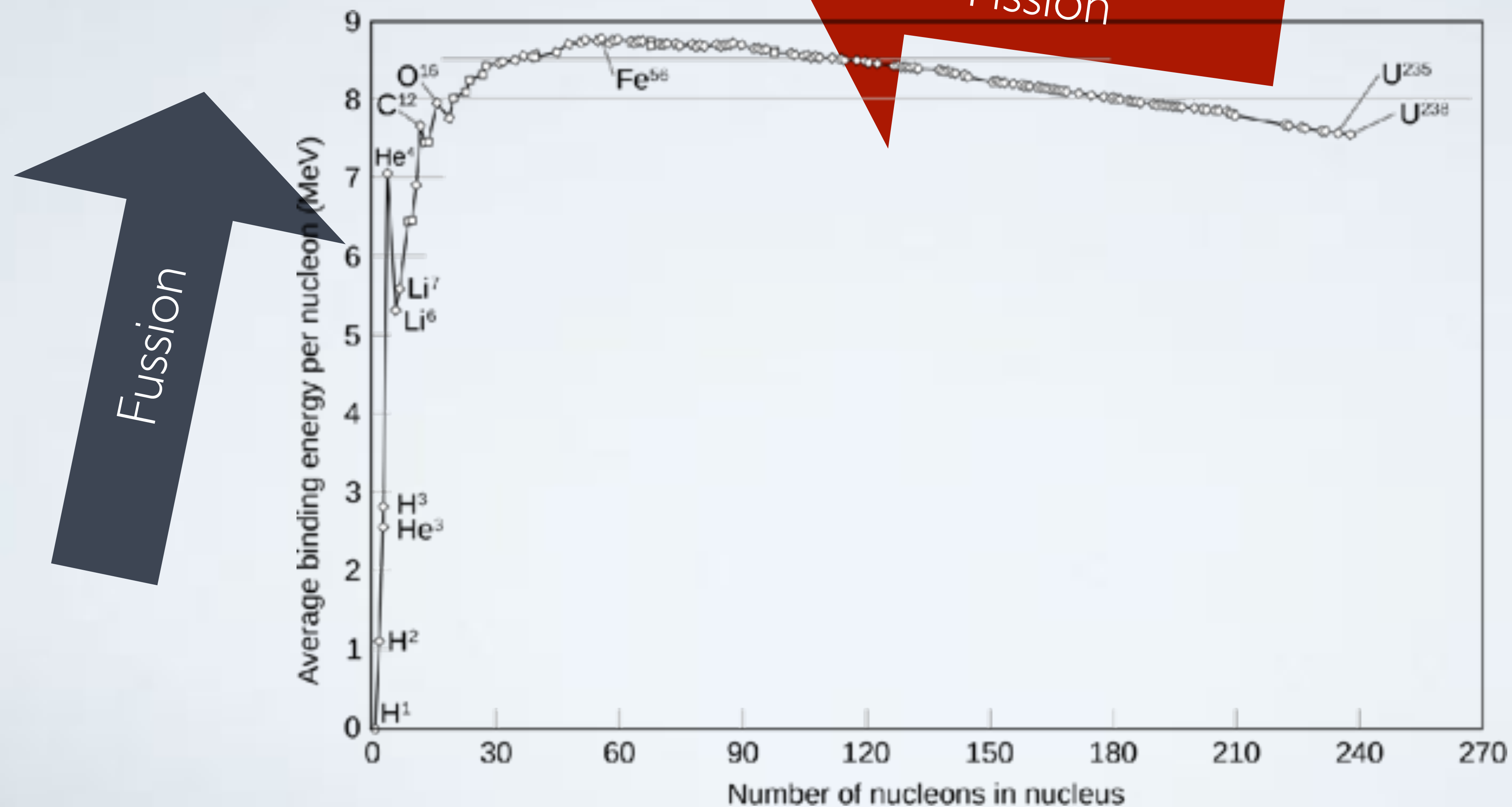
$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

- Kvantemekanisk tryk:

$$P_{\text{deg}}(r) = \begin{cases} (3\pi^2)^{\frac{2}{3}} \frac{\hbar^2}{5m} \left(\frac{N_{e,n}}{V} \right)^{\frac{5}{3}} & \text{ikke-relativistisk} \\ \frac{1}{4} (3\pi^2)^{\frac{1}{3}} \hbar c \left(\frac{N_{e,n}}{V} \right)^{\frac{4}{3}} & \text{ultra-relativistisk} \end{cases}$$

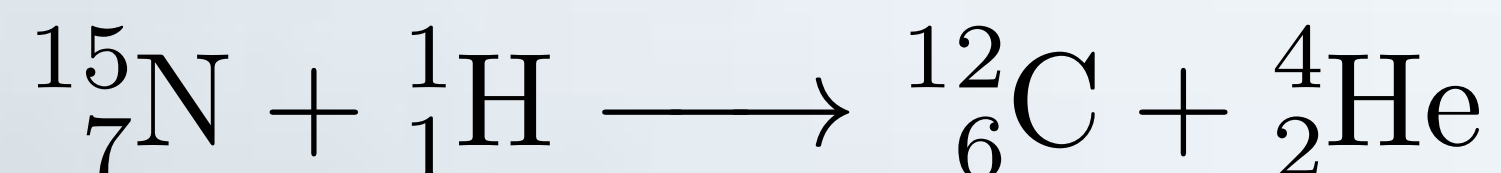
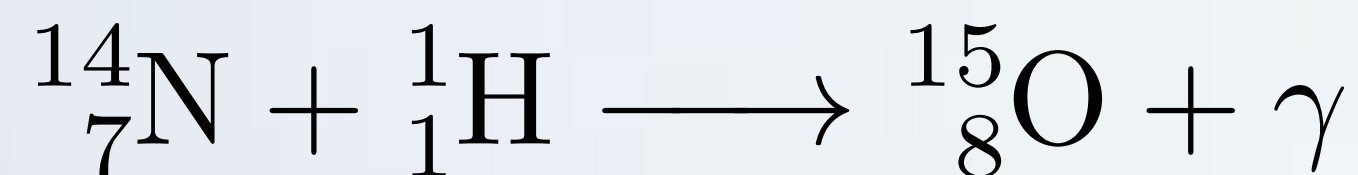
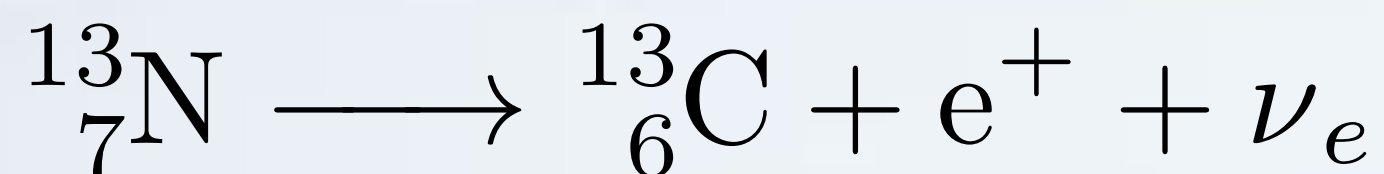
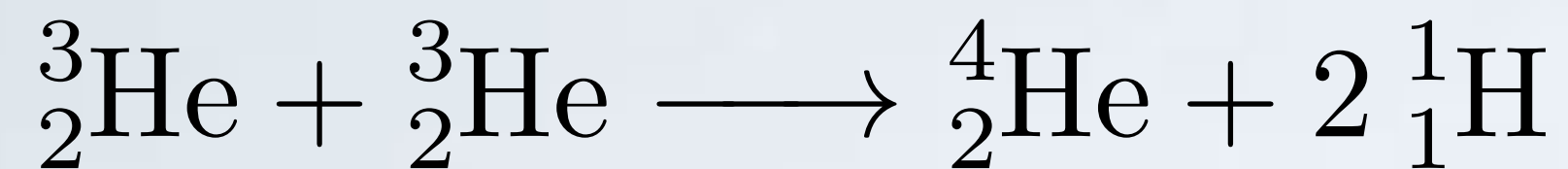
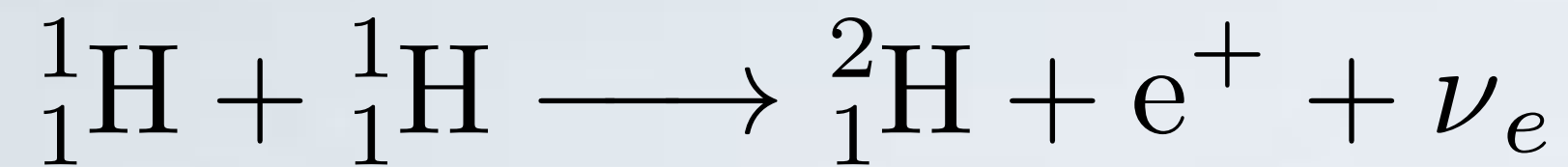


HVOR KOMMER TEMPERATUREN FRA?



HYDROGENFUSSION

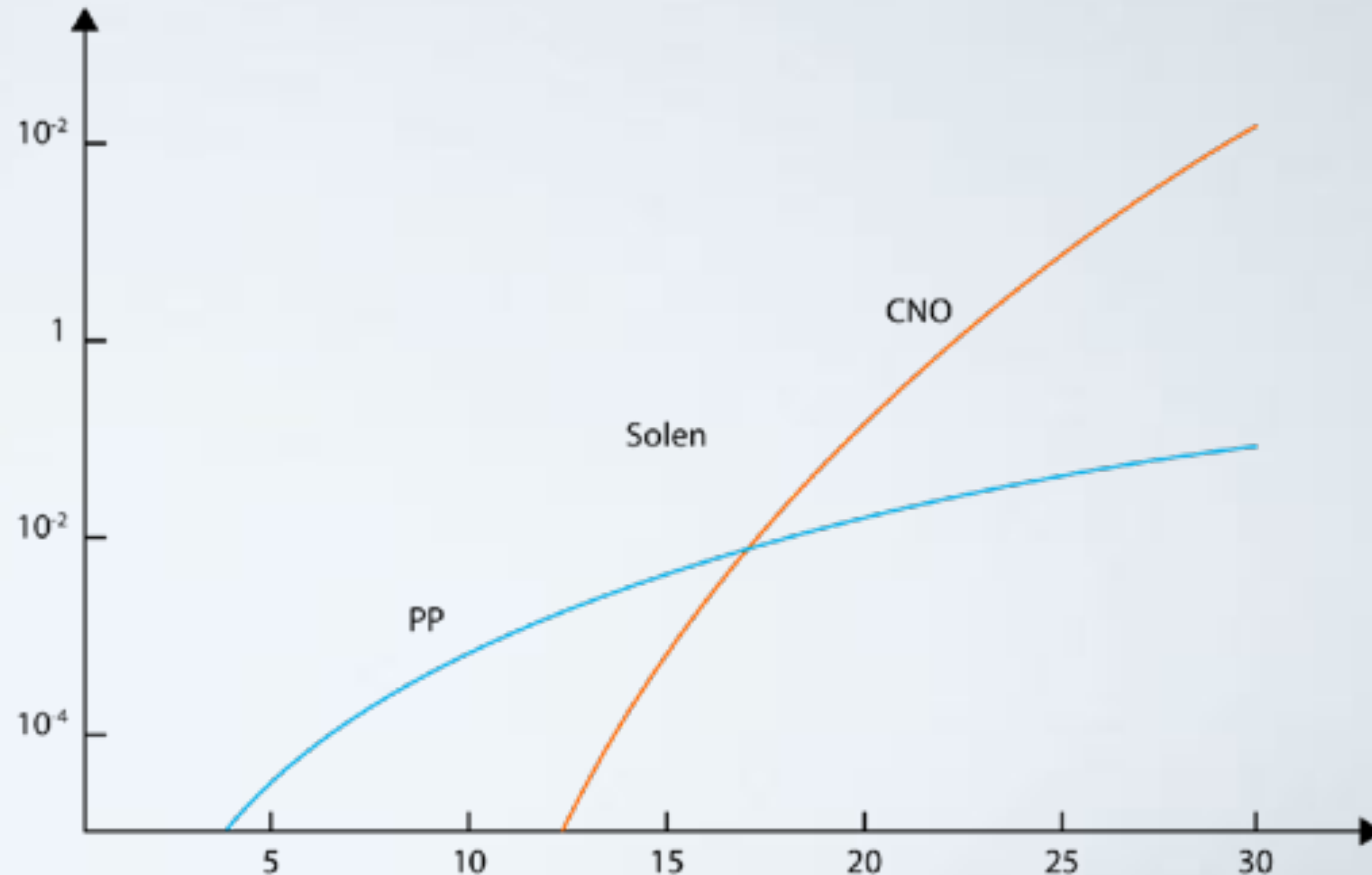
pp-I



CNO

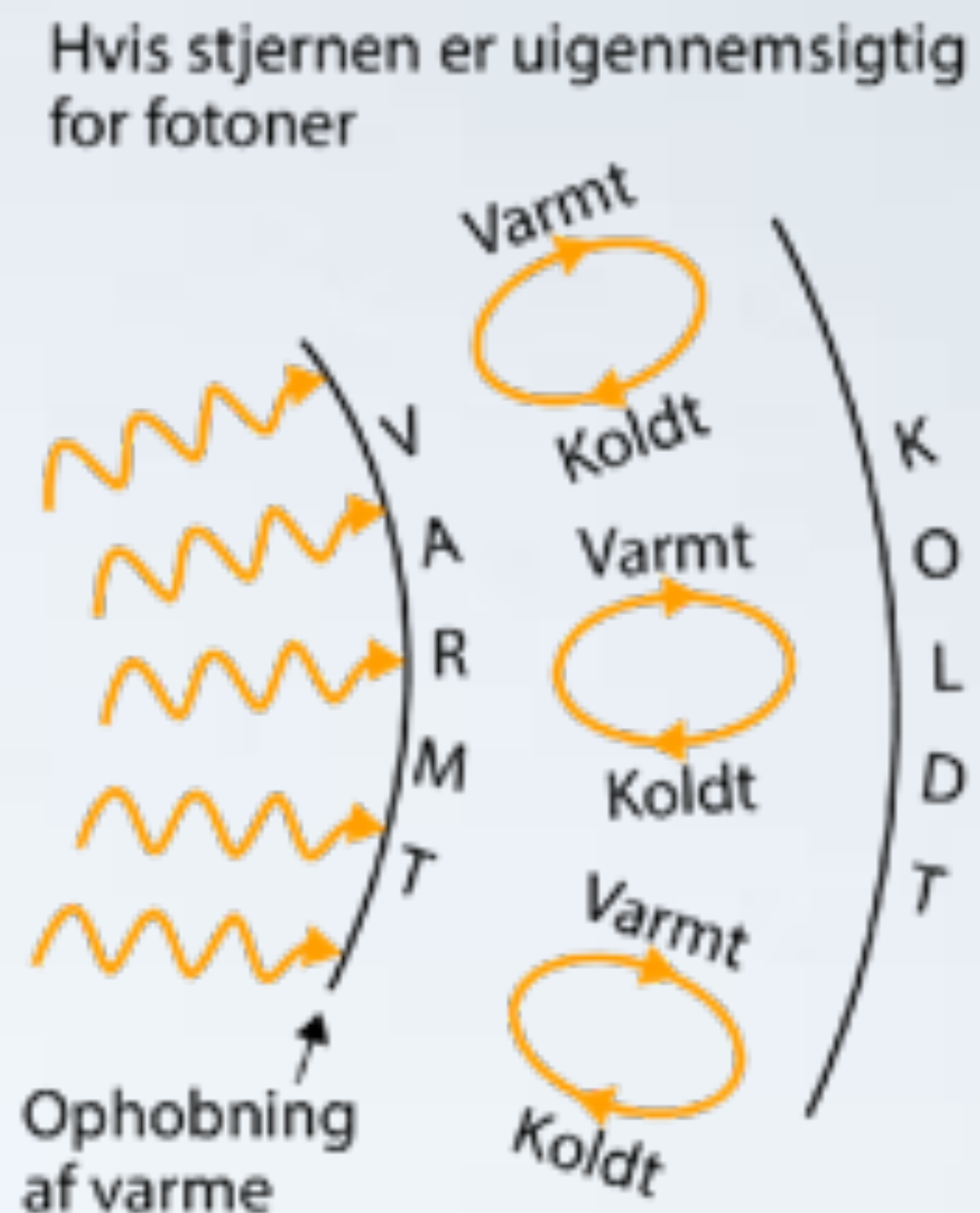
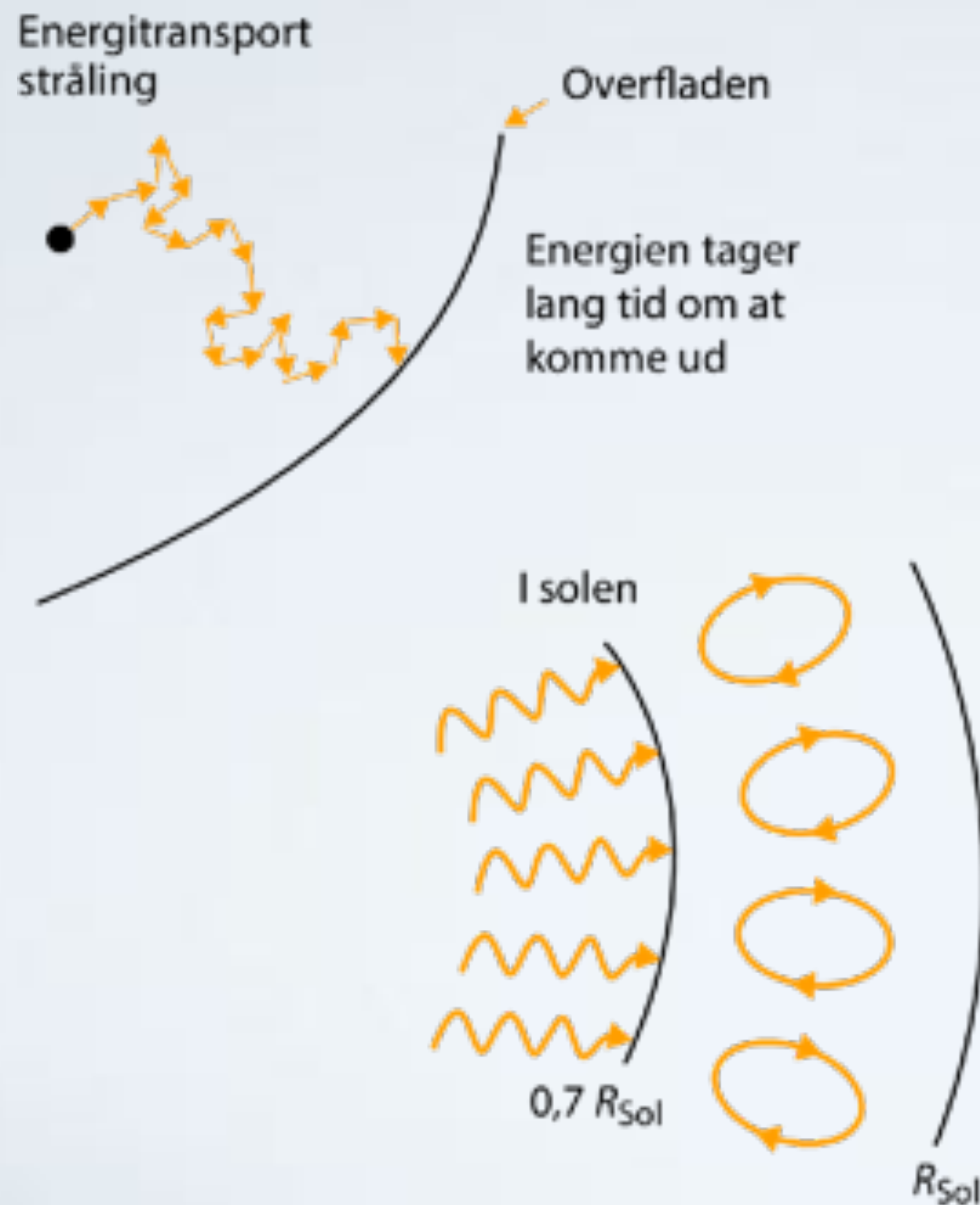
Energiproduktion

W/kg



Kernetemperatur i stjernen (10⁶ K)

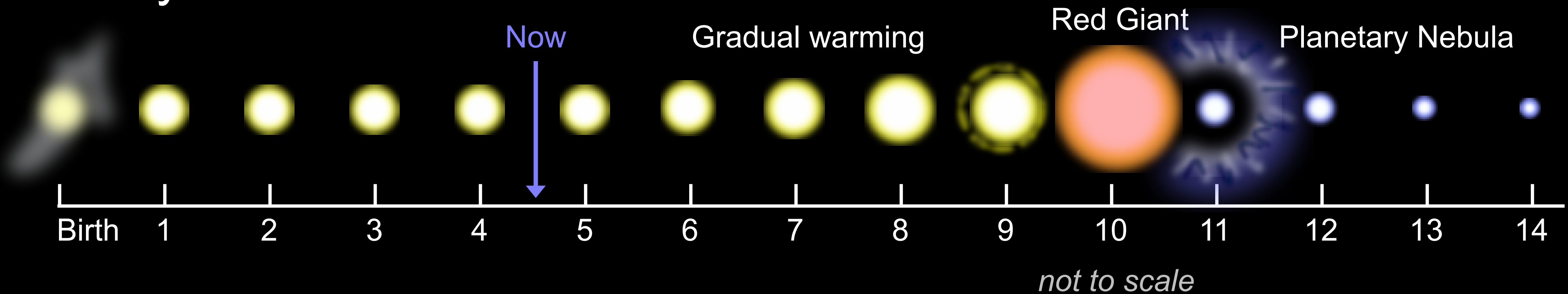
HVORDAN KOMMER ENERGIE UD?



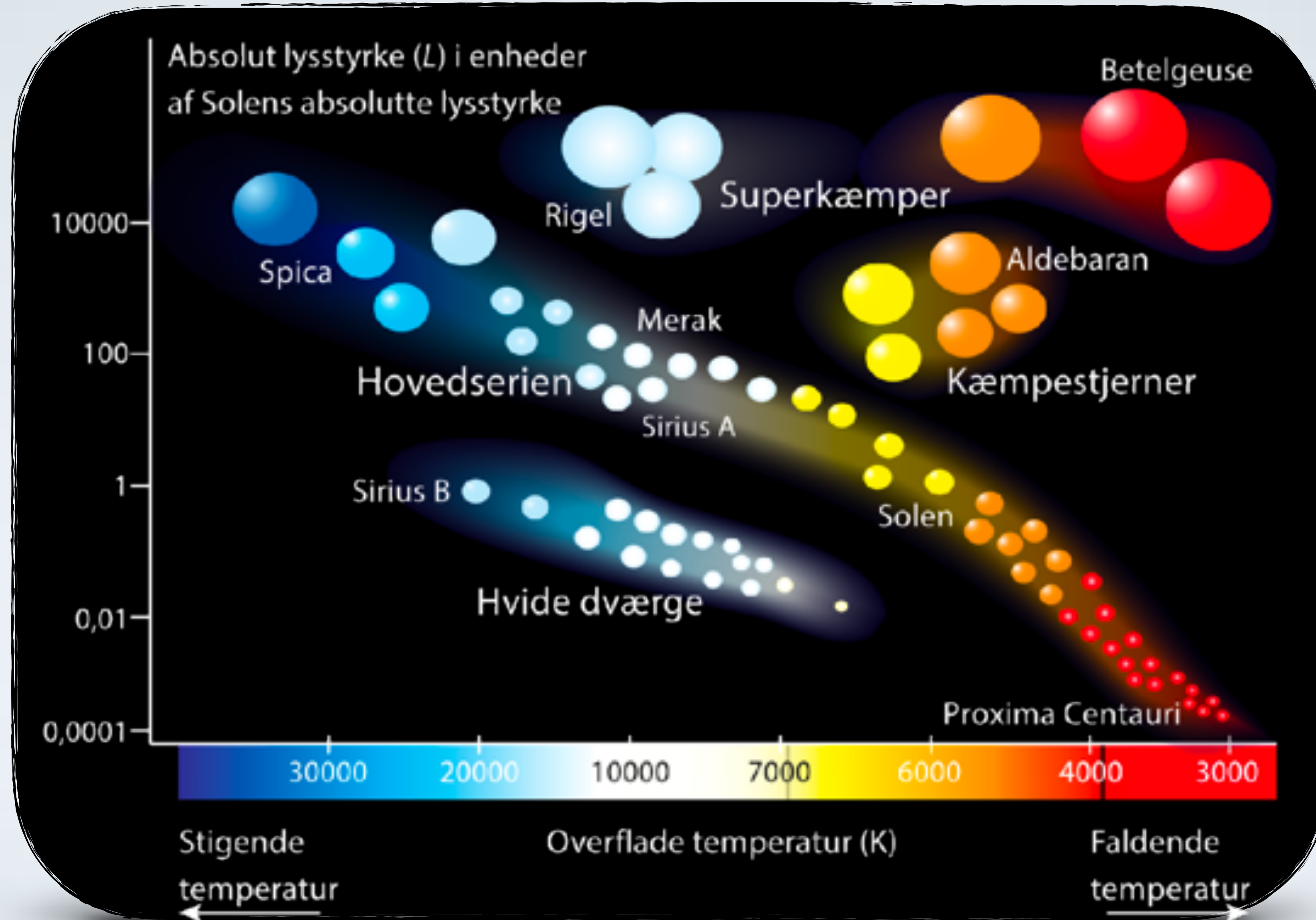
STJERNEUDVIKLING

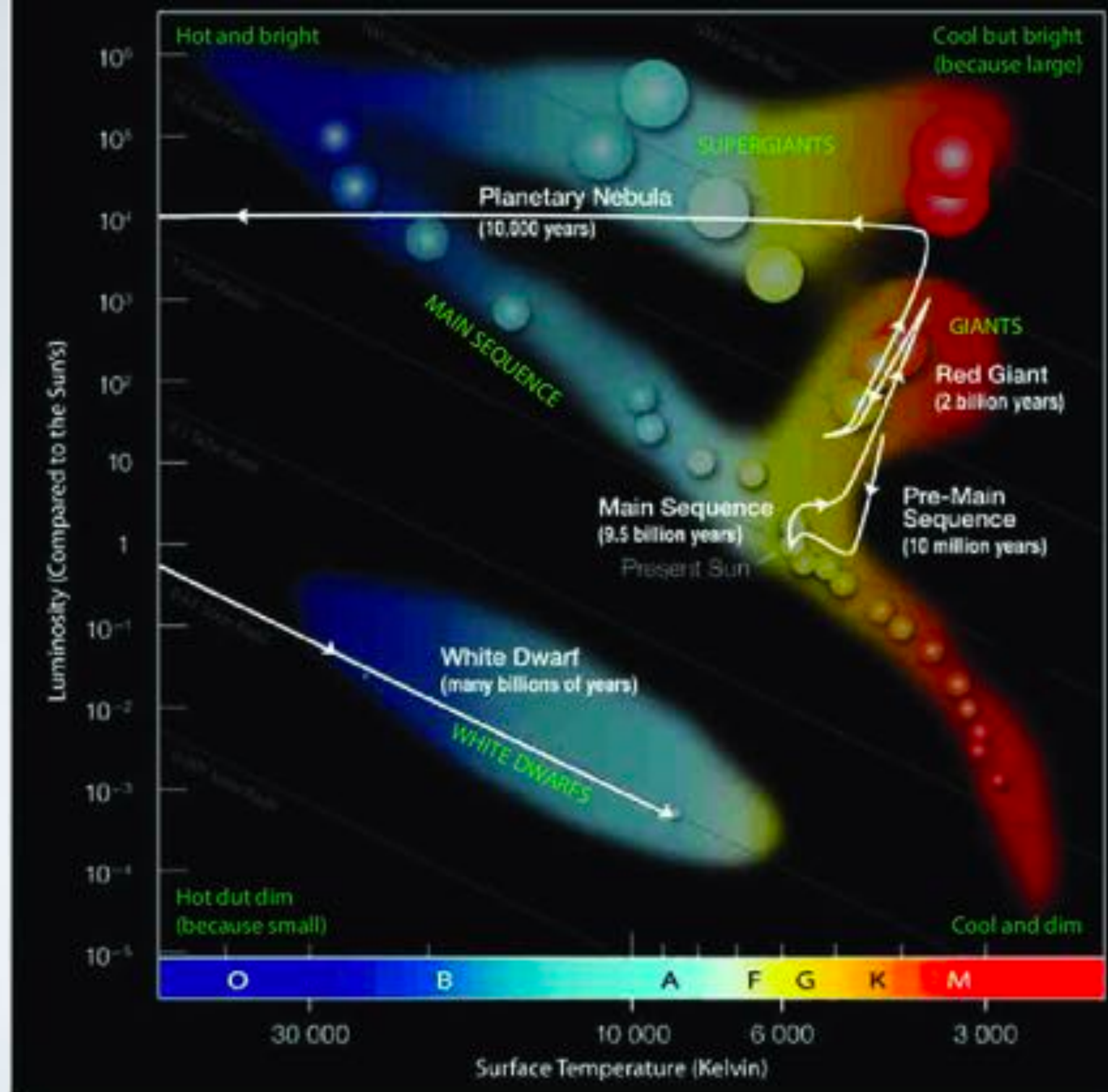
SOLENS UDVIKLING

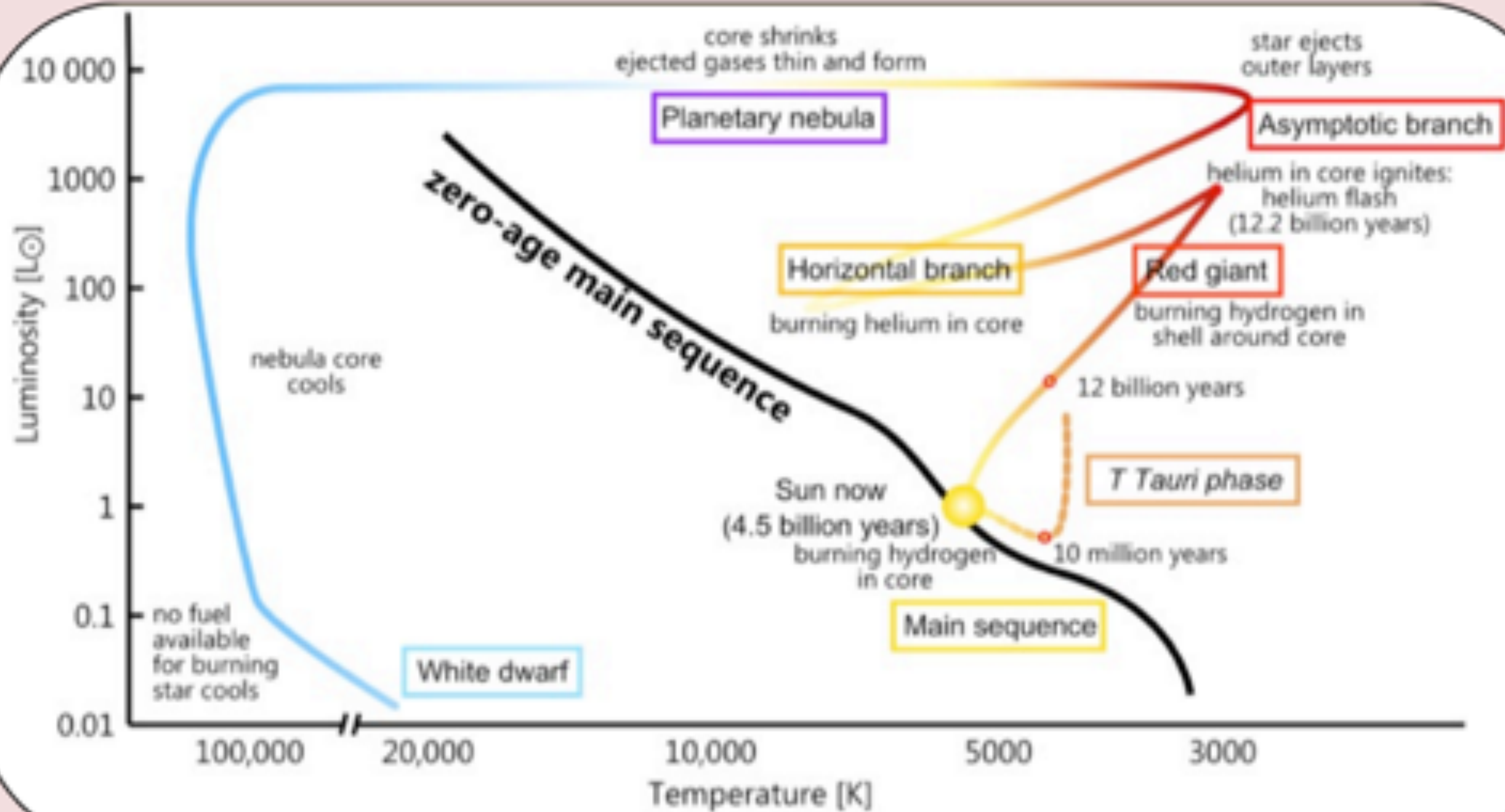
Life Cycle of the Sun

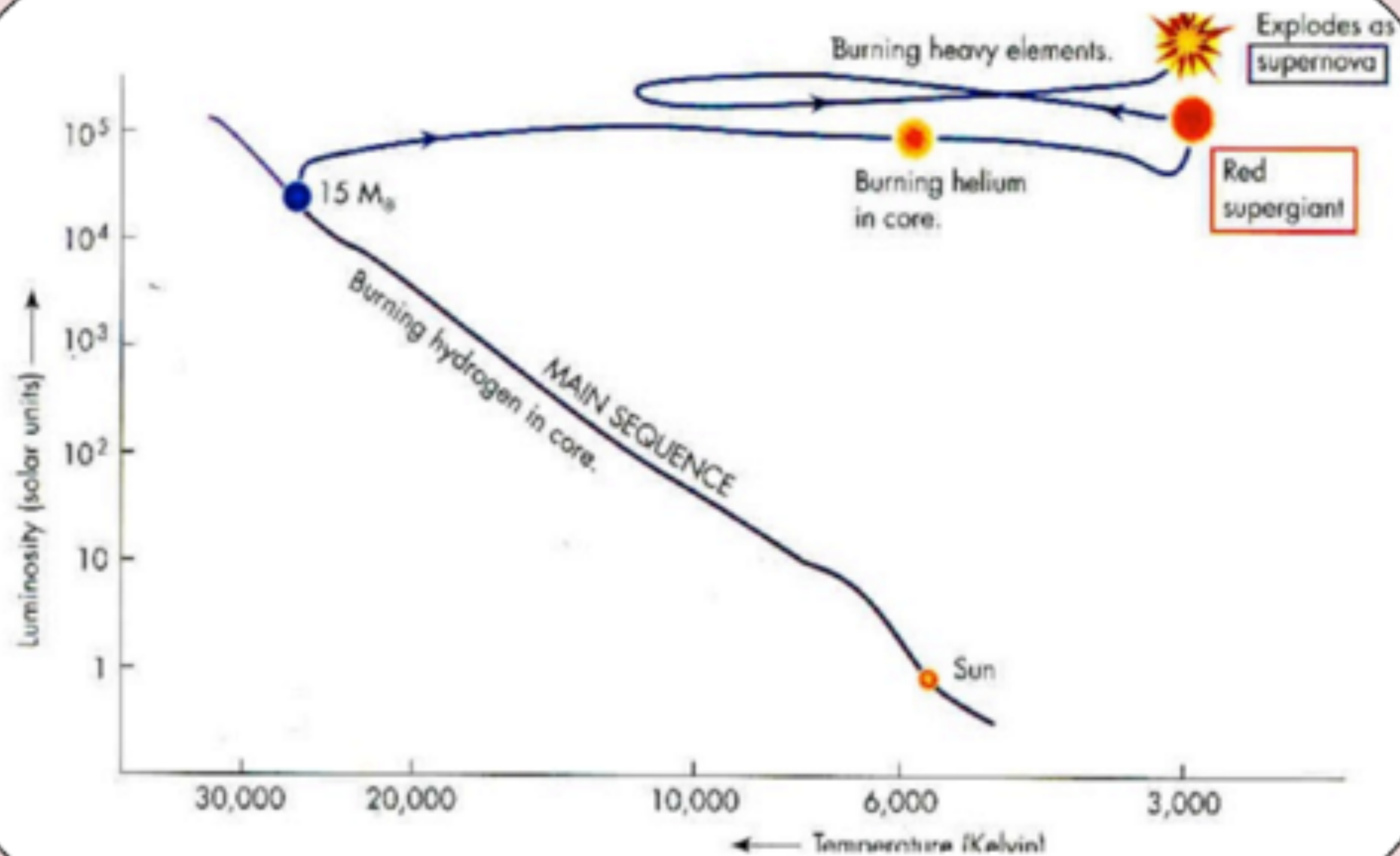


HERTZSPRUNG-RUSSEL DIAGRAMMET



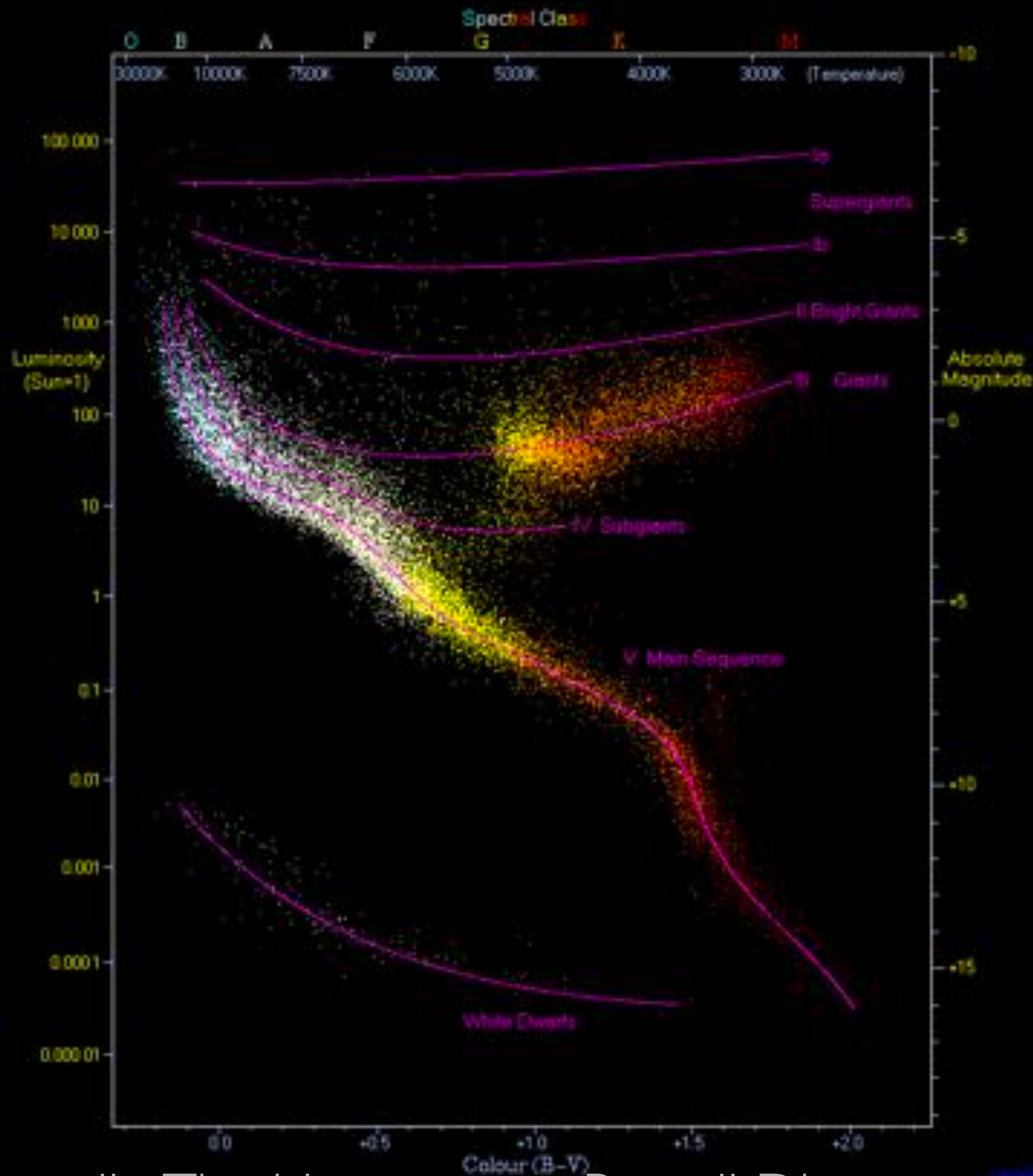






OBSERVATIONELT

HR-DIAGRAM



By Richard Powell - The Hertzsprung Russell Diagram

Masse ($M_{\odot}$)	Levetid (Giga-år)
60	0.003
30	0.011
10	0.032
3	0.370
1.5	3
1	10
0.1	1000

$140M_{\odot} \lesssim M \lesssim 250M_{\odot}$
(Pair-instability)

$90M_{\odot} \lesssim M \lesssim 140M_{\odot}$

$40M_{\odot} \lesssim M \lesssim 90M_{\odot}$

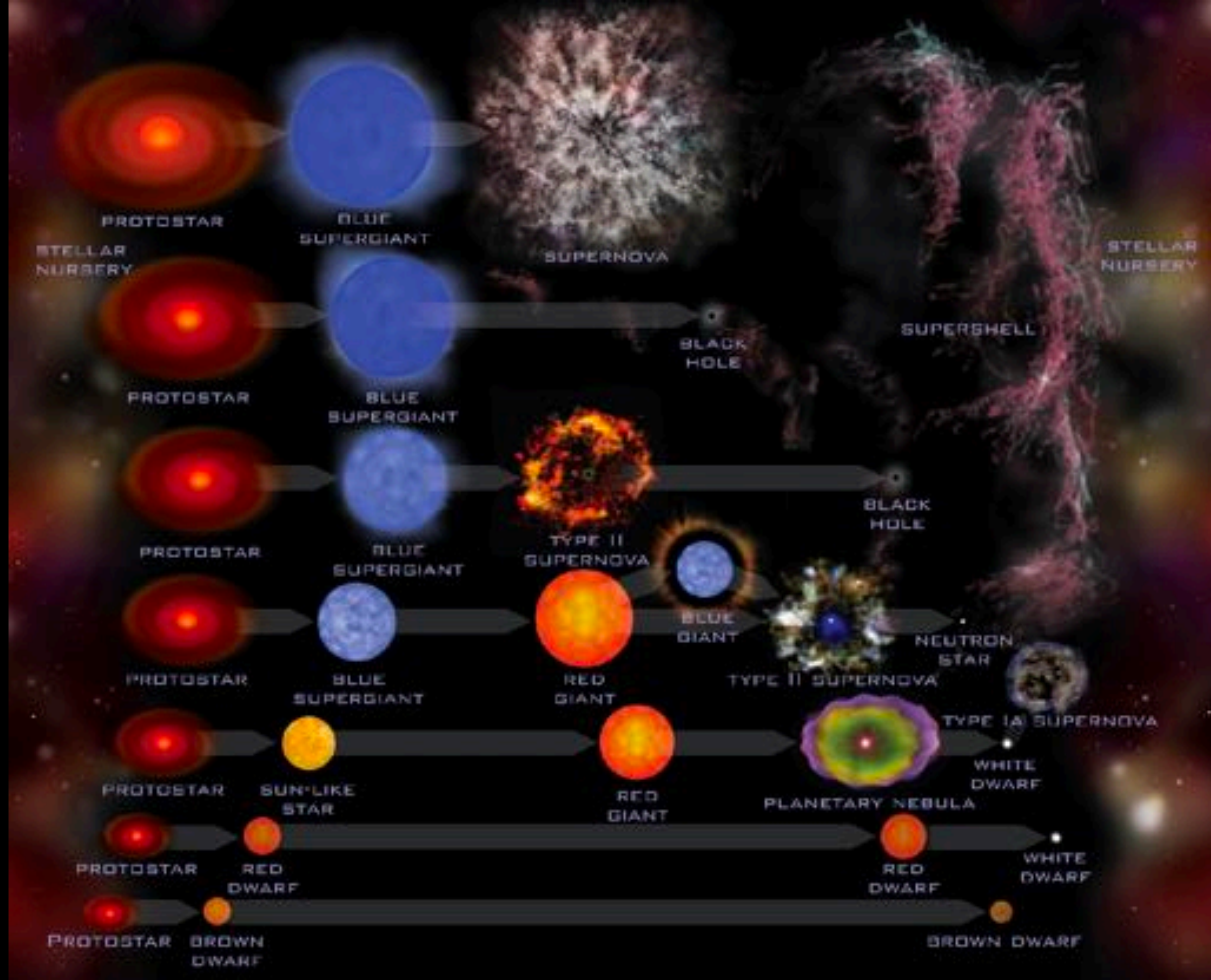
$9M_{\odot} \lesssim M \lesssim 40M_{\odot}$

$2M_{\odot} < M \lesssim 9M_{\odot}$

$0.5M_{\odot} < M \lesssim 2M_{\odot}$

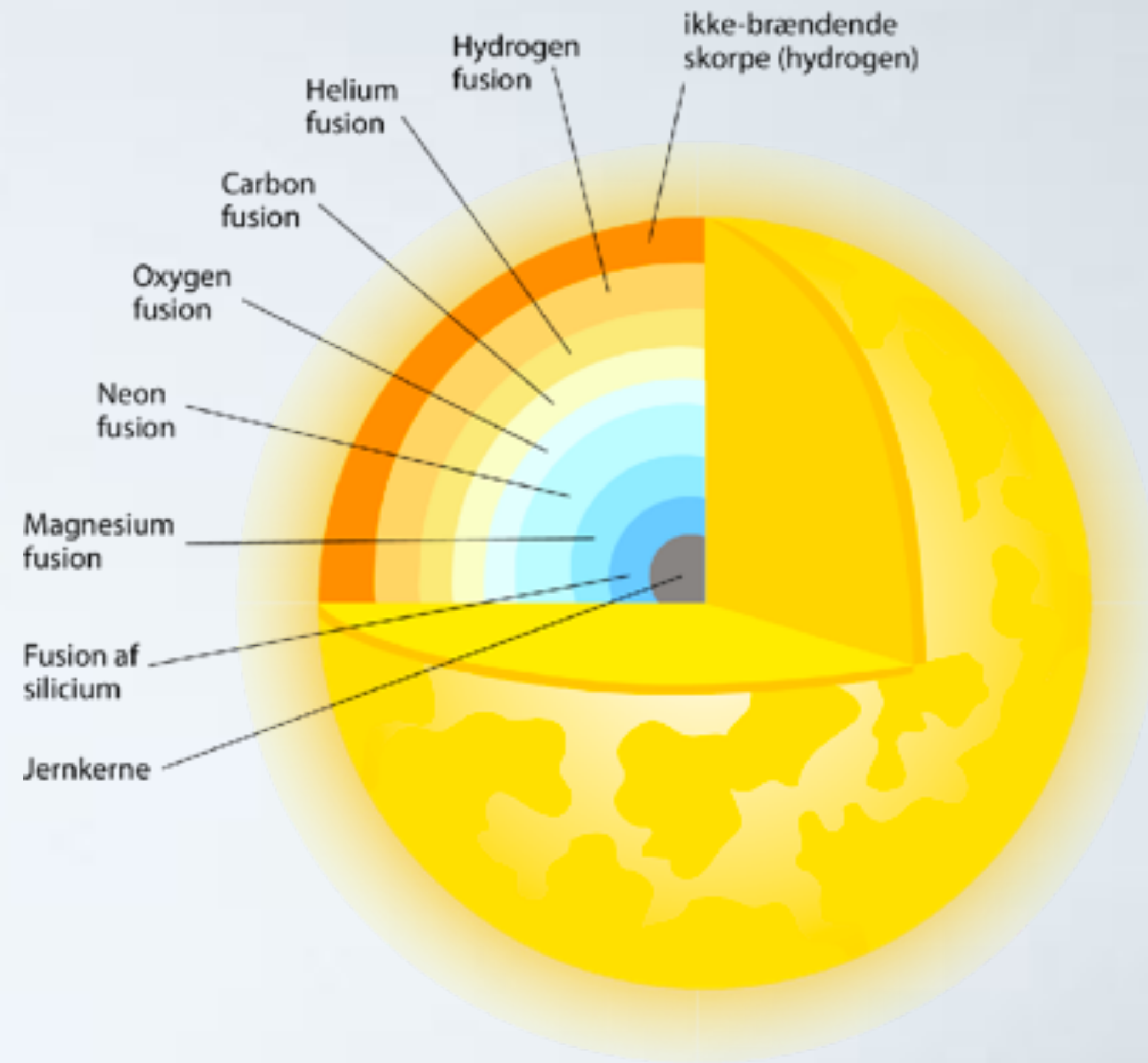
$0.08M_{\odot} < M < 0.5M_{\odot}$

$0.0125M_{\odot} < M < 0.08M_{\odot}$



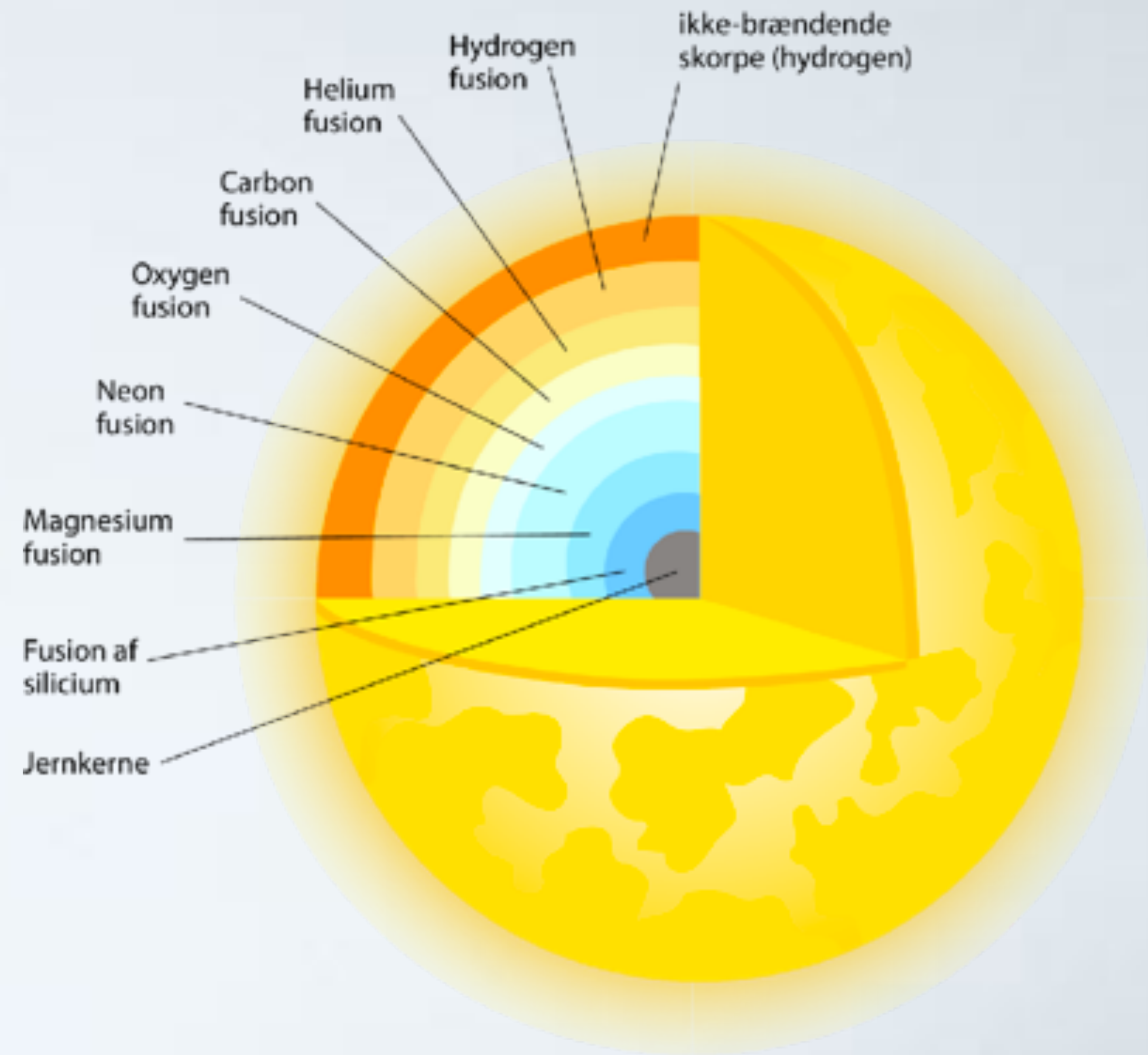
STJERNERS ENDELIGT

- Rød kæmpe → Hvid dværg
Stabiliseret af degenereret elektron-gas
- Neutronstjerne
Stabiliseret af degenereret neutron-gas
9-25+ $M_{\odot}$
- Sort hul 25+ $M_{\odot}$
Ikke stabiliseret

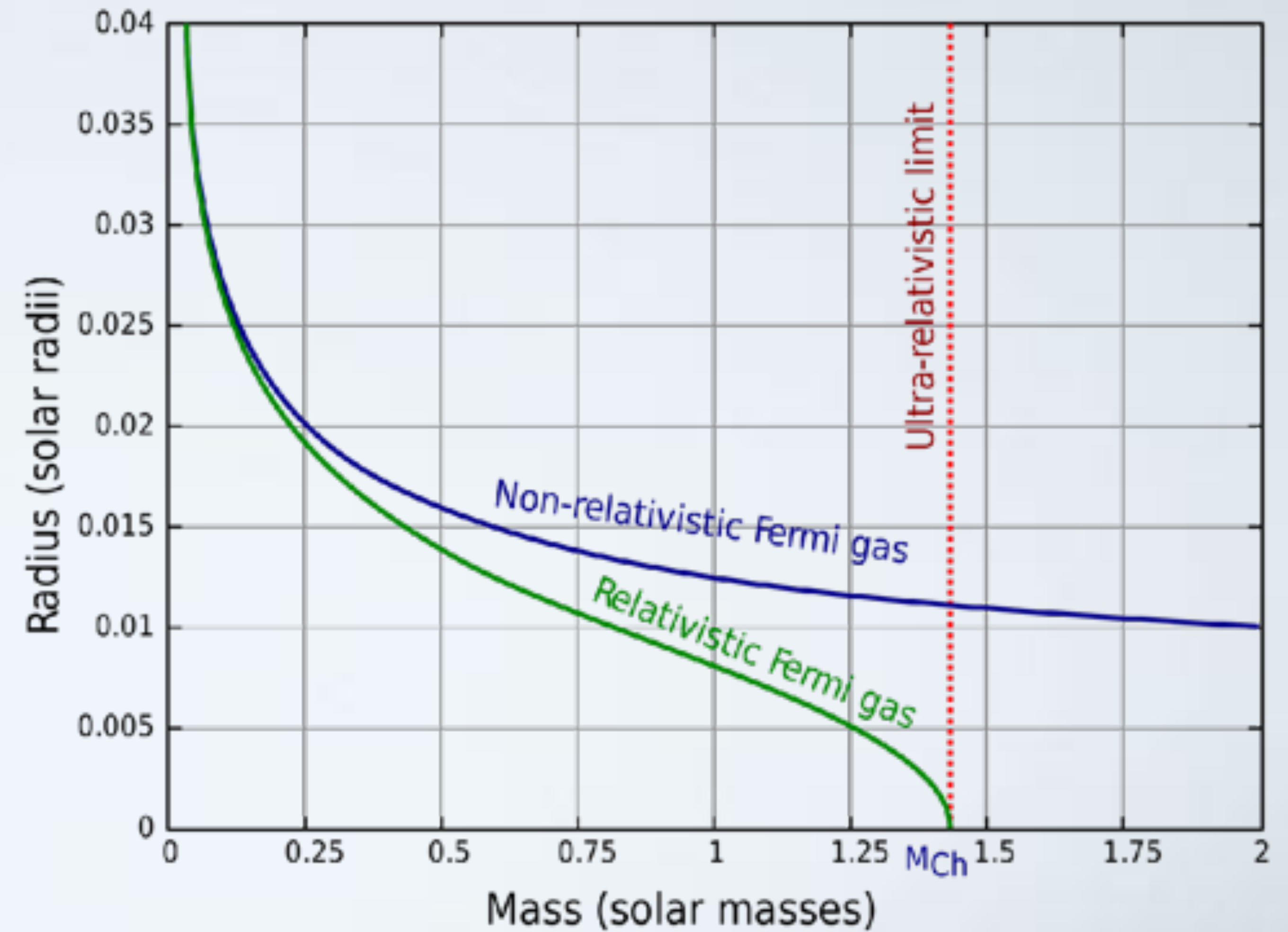


STJERNERS ENDELIGT

- Rød kæmpe → Hvid dværg
Stabiliseret af degenereret elektron-gas
- Neutronstjerne
Stabiliseret af degenereret neutron-gas
9-25+ $M_{\odot}$
- Sort hul 25+ $M_{\odot}$
Ikke stabiliseret



HVIDE DVÆERGE



NEUTRONSTJERNER

